$$p_\theta(x_0) = \int p_\theta(x_{0:T})\, dx_{1:T}$$ → marginal distribution & Markov property

$$-\log p_\theta(x_0) = -\log \int p_\theta(x_{0:T})\, dx_{1:T}$$

$$= -\log \int p_\theta(x_{0:T})\, dx_{1:T} \cdot \frac{q(x_{1:T}|x_0)}{q(x_{1:T}|x_0)}$$ → Same as VAE..

$$\geq E_{x \sim q(x_{1:T}|x_0)}\left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]$$

$-E_{q(x_0)}[\log p_\theta(x_0)]$ is the exact objective (expectation of all data)

we can change to $E_{q(x_{0:T})}[\quad]$

$$L = \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = E_q\left[-\log \frac{p_\theta(x_T)\cdot \prod_{1}^{T} p_\theta(x_{t-1}|x_t)}{\prod_{1}^{T} q(x_t|x_{t-1})}\right]$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \quad \cdots ①$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right] \quad \cdots ②$$

$$= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} \cdot \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} - \log \frac{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)}\right] \quad \cdots ③$$

$$= \mathbb{E}_q\left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right] \quad \cdots ④$$

① → ② : Σ expansion

② → ③ : $$q(x_t|x_{t-1}) = q(x_{t-1}|x_t)\cdot \frac{q(x_t)}{q(x_{t-1})}$$

$$= q(x_{t-1}|x_t, x_0)\cdot \frac{q(x_t|x_0)}{q(x_{t-1}|x_0)}$$ due to Markov property

③ → ④ : $$\sum \log \frac{q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

$$= \frac{q(x_1|x_0)\cdot \cancel{q(x_2|x_0)} \cdots}{\cancel{q(x_2|x_0)}\cdot \cancel{q(x_3|x_0)} \cdots \quad q(x_T|x_0)}$$

$$= \frac{q(x_1|x_0)}{q(x_T|x_0)}$$

$$\mathcal{L} = \mathbb{E}_q\left[D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\,\|\,p(\mathbf{x}_T)) + \sum_{t>1} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

⟨Denoising term⟩

$D_{KL}\left(q(x_{t-1}|x_t) \,||\, p_\theta(x_{t-1}|x_t) \right)$

$\hookrightarrow q(x_{t-1}|x_t, x_0) = q(x_t|x_{t-1}, x_0) \cdot \dfrac{p(x_{t-1}|x_0)}{p(x_t|x_0)}$

① ② ③

① $\sim N(\sqrt{\alpha_t}x_0, (1-\alpha_t)I)$

② $\sim N(\sqrt{\bar{\alpha}_{t-1}}x_0, (1-\bar{\alpha}_{t-1})I)$

③ $\sim N(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$

$\Rightarrow q(x_{t-1}|x_t, x_0) = N(\tilde{\mu}, \tilde{\sigma}_t^2 I)$

where $\tilde{\mu}(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$ and $\tilde{\sigma}_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$.

So, KL of two Normal distribution with same variance $\tilde{\sigma}_t^2$?

$\rightarrow \frac{1}{2\tilde{\sigma}_t^2}|\mu_p - \mu_q|^2$

We can change mu predictor to x_0 or ϵ_t using ...

$x_0 \rightarrow \quad \tilde{\mu}(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$

$\epsilon_t \rightarrow \begin{cases} x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon \quad \rightarrow \quad x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon) \\ \tilde{\mu}(x_t, x_0) \xleftarrow{\text{Substitute}} \end{cases}$

$\Rightarrow \tilde{\mu}(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta \right)$

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t)$$
$$= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right)$$

< DDIM >

Let's assume that ① reverse process does not follow Markov chain

② $q_\sigma(x_{t-1} | x_t, x_0)$ can be expressed with x_0, x_T's linear combination

$$\Rightarrow q_\sigma(x_{t-1}|x_t, x_0) = \mathcal{N}(w_0 x_0 + w_T x_T, \sigma_T^2 I)$$

note that $q_\sigma(x_{t-1}|x_0) = \int q_\sigma(x_{t-1}|x_t, x_0)\, q_\sigma(x_t|x_0)\, dx_t$

$p(x) = \mathcal{N}(\mu, \sigma_1^2 I) \rightarrow x = \mu + \sigma_1 \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$

$p(y|x) = \mathcal{N}(ax+b, \sigma_2^2 I) \rightarrow z = ax+b+\sigma_2\epsilon \Rightarrow E[z] = E[ax+b+\sigma_2\epsilon] = aE[x]+b = \underline{a\mu+b}$

$Var[z] = Var[ax+b+\sigma_2\epsilon] = a^2 Var[x] + \sigma_2^2 Var[\epsilon]$

$= \underline{a^2\sigma_1^2 + \sigma_2^2}$

$\Rightarrow p(y) = \mathcal{N}(a\mu+b, (\sigma_2^2 + a^2\sigma_1^2) I)$

Apply these technique above ...

$$q_\sigma(x_{t-1}|x_t, x_0) = \mathcal{N}(w_0 x_0 + w_T x_T, \sigma_T^2 I)$$
$$q_\sigma(x_t | x_0) = \mathcal{N}(\sqrt{\bar\alpha_t}\, x_0, (1-\bar\alpha_t) I)$$

$$\Rightarrow q_\sigma(x_{t-1}|x_0) = \mathcal{N}\left(\underline{w_0 x_0 + w_t\sqrt{\bar\alpha_t}\, x_0}, \underline{((1-\bar\alpha_t) w_T^2 + \sigma_t^2) I}\right)$$
$$= \mathcal{N}(\underline{\sqrt{\bar\alpha_{t-1}}\, x_0}^{①}, \underline{(1-\bar\alpha_{t-1}) I})^{②}$$

② $\Rightarrow$ $w_t = \sqrt{\dfrac{1-\bar\alpha_{t-1}-\sigma_t^2}{1-\bar\alpha_t}}$

$w_0 = \sqrt{\bar\alpha_{t-1}} - \sqrt{\bar\alpha_t} \cdot \sqrt{\dfrac{1-\bar\alpha_{t-1}-\sigma_t^2}{1-\bar\alpha_t}}$

$$\therefore q_\sigma(x_{t-1}|x_0, x_t) = \mathcal{N}\left(\sqrt{\bar\alpha_{t-1}}\mathbf{x}_0 + \sqrt{1-\bar\alpha_{t-1}-\sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar\alpha_t}\mathbf{x}_0}{\sqrt{1-\bar\alpha_t}}, \sigma_t^2 \mathbf{I}\right)$$

Same training step with DDPM, use $q_\sigma(x_{t-1}|x_t, x_0)$ instead of $p_\theta(x_{t-1}|x_t)$ when we conduct sampling

< CFG >

- From Classifier Guidance ...

$$\nabla_{x_t} \log p(x_t, y) = \underbrace{\nabla_{x_t} \log p(y|x_t)}_{\text{classifier result}} + \nabla_x \log p(x_t)$$

$\rightarrow$ classifier result

$\rightarrow p(x_t) = \mathcal{N}(\sqrt{\bar\alpha_t}\, x_0, (1-\bar\alpha_t)I)$

$\rightarrow \nabla_{x_t} \log p(x_t) = \nabla_{x_t} \log \left(\frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{|x_t - \sqrt{\bar\alpha_t}\, x_0|^2}{2\sigma}\right)\right)$

$= -\frac{1}{\sigma}(x_t - \sqrt{\bar\alpha_t}\, x_0)$

$= -\frac{1}{\sqrt{1-\bar\alpha_t}} \epsilon_\theta(x_t, t)$

$$= -\frac{1}{\sqrt{1-\bar\alpha_t}} \left(-\sqrt{1-\bar\alpha_t}\, \nabla_{x_t} \log p(y|x_t) + \epsilon_\theta(x_t, t)\right)$$

$$= -\frac{1}{\sqrt{1-\bar\alpha_t}} \tilde{\epsilon}_\theta(x_t, t, y)$$ $\rightarrow$ by meaning of score function

if we put weight w on $\nabla_{x_t} \log p(y|x_t)$ term.

$$\rightarrow \nabla_{x_t} \log p(x_t, y) = \epsilon_\theta(x_t, t) - w\sqrt{1-\bar\alpha_t}\, \nabla_{x_t} \log p(y|x_t)$$

$$= -\sqrt{1-\bar\alpha_t}\, \nabla_{x_t} \log p(x_t) - w\sqrt{1-\bar\alpha_t}\, \nabla_x \log p(y|x_t)$$

$$= -\sqrt{1-\bar\alpha_t}\, \nabla_{x_t} \log \left(p(x_t)\, p(y|x_t)^w\right)$$

$\hookrightarrow p(x_t)\, p(y|x_t)^w$

$$= p(x_t) \left(\frac{p(x_t|y)\, p(y)}{p(x_t)} \right)^w$$

$$\propto \underbrace{p(x_t)}_{\epsilon_\theta(x_t, t)} \underbrace{\left(\frac{p(x_t|y)}{p(x_t)} \right)^w}_{w(\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t))}$$

$$\propto w\, \epsilon_\theta(x_t, t, y) + (1-w)\, \epsilon_\theta(x_t, t, \emptyset)$$

$$\doteqdot (w+1)\, \epsilon_\theta(x_t, t, y) + \epsilon_\theta(x_t, t, \emptyset)$$